Task 4, Iteration 1: Codifying Idea 1.1

BITS F471 Group 3 ·

In short: we tested idea 1.1 on GSM8K math problems, and it did not work. Picking answers with an AR model was worse than a simple majority vote. The best rule we found matched majority vote but did not beat it.

1. The idea

Diffusion LMs (DLMs) such as DiffuLLaMA, our anchor paper's model, are made by converting ordinary left-to-right (AR) models. Idea 1.1 asks whether what the AR model knew survives that conversion and can still help.

1. Measure how similar each DLM's inner layers still are to its original AR model (the CKA score).
2. Let the DLM write 10 answers per question, and let an AR model pick the one it finds most likely.
3. Check whether DLMs that stayed more similar to their AR model gain more from step 2.

We chose this idea from our top five because it is the only one that produces a task accuracy we can compare with baselines, and it needs no training. That fits our compute: one shared workstation.

2. What we did

- **Models:** five DLMs. DiffuGPT-S, DiffuGPT-M and DiffuLLaMA come from the anchor paper. Dream-7B was converted from Qwen2.5-7B. LLaDA-8B was trained from scratch, so it has no AR original; it is our control.
- **Task:** GSM8K math word problems. Each model wrote 10 answers per question, with 4 worked examples in the prompt and 64 denoising steps.
- **Answer pickers:** five AR models: GPT-2, Qwen2.5-0.5B, TinyLlama, and each DLM's own 7B original (Qwen2.5-7B for Dream, Llama-2-7B for DiffuLLaMA). The picker chooses the answer it finds most likely.
- **Similarity:** CKA between each DLM and its AR original, layer by layer, on 2,000 short passages of Wikipedia text.

- **Baselines:** the first answer, majority vote (the most common final answer, also called self-consistency), and the best possible pick.

Codifying with an LLM. We used Arbor, one of the four reference systems and the only one with complete public code. Arbor is an AI research agent: it proposes a hypothesis, writes it as code (here, the rule that picks one of the 10 answers), tests it, and records what it learned in a tree of ideas. It keeps a change only if it also wins on questions it was not tuned on. We ran it with a local Qwen3.8-27B model and reviewed every proposal before it ran.

Fair test. The agent tuned on 200 questions and checked its changes on 200 more. The final scores below use 300 further test questions it never saw.

3. How we steered the agent

The brief asks us to refine the idea by giving feedback to the model. Every Arbor proposal came to us first, and we answered each one in writing.

- **New goal.** Our first trials showed AR picking loses to majority vote, so we set Arbor's goal as beating majority vote, not the first answer.
- **Rejected twice.** Arbor twice proposed copying other models' answers to the same question. We rejected it: that mixes models, is unfair to every single-model baseline, and makes the score depend on question order.
- **Redirected.** We pointed it at a signal inside each answer instead: re-check every arithmetic step, and trust answers with a mistake less. Arbor built this rule and kept it.
- **Corrected.** It called Llama-2 LLaDA's original model; LLaDA has none.

In 68 minutes Arbor tested 4 ideas and kept 1.

4. Results

Accuracy (%) on the 300 unseen GSM8K test questions:

Model	First answer	Majority vote	Best AR pick	Agent's rule	Best possible*
DiffuGPT-S	1.7	2.3	2.0	2.3	14.7
DiffuGPT-M	0.7	1.3	2.3	1.3	8.7
DiffuLLaMA	0.3	2.0	2.3	2.0	9.7
Dream-7B	41.0	**73.3**	32.0	73.0	88.0

Model	First answer	Majority vote	Best AR pick	Agent's rule	Best possible*
LLaDA-8B	48.7	**73.0**	52.0	73.0	88.3

*Best possible: the question counts as right if any of the 10 answers is right.

- **AR picking does not help.** On Dream and LLaDA it is 21 to 41 points worse than majority vote, even with each DLM's own 7B original as the picker. On Dream it is worse than just taking the first answer.
- **Majority vote is strong.** It closes about two-thirds of the gap between the first answer and the best possible (Dream: from 41 to 73, out of 88).
- **The agent's best rule only ties majority vote.** On the unseen questions it scores 0.3 points lower on Dream and the same on LLaDA ($p = 1.0$ for both). It looked 1.5 points better on the questions it was tuned on, but that did not hold.
- **The three small models get almost everything wrong**, so they tell us little.

Does similarity predict the gain? The table shows each model's CKA similarity to its AR original, and how much of the possible gain a Qwen2.5-7B picker achieved (500 test questions).

Model	CKA similarity (0 to 1)	Gain from AR picking (% of possible)
DiffuGPT-S	0.89	+5.3
DiffuGPT-M	0.87	+10.0
DiffuLLaMA	0.75	0.0
Dream-7B	0.51	−13.9
LLaDA-8B	none (no AR original)	−8.4

The trend roughly matches the idea's prediction: more similar models gain more. But this is not evidence. With 4 models the trend is not significant (best $p = 0.083$). The small models move by one or two questions, and LLaDA, which has no AR original, also loses.

5. What it means, and next steps

Idea 1.1 as written does not beat the baseline. Majority vote, which is also the anchor paper's own baseline, is still the best method we have.

In the next iteration we will:

1. **Close the gap to published numbers.** Our models score below their published results (Dream 41% vs 77%, LLaDA 49% vs 70%). We will rerun one answer per question at the published settings, to see how much comes from our cheaper settings (fewer steps, more randomness) and rule out a bug.
2. **Try new signals for picking answers.** In-answer rules seem used up. Next we try the DLM's own confidence in each answer, and an answer checker.
3. **Compare picking with more steps.** The idea also asks whether picking a better answer beats simply running the model for more steps. We will run 16 and 32 steps and compare at equal compute.
4. **Compare with published work.** Add a published answer-picking method for DLMs as the state-of-the-art comparison.
5. **A stronger agent run.** Start Arbor from majority vote, so anything it keeps truly beats it, and borrow safeguards from the other reference papers: written rules the agent must not break (AutoSOTA), a shared memory of what failed (DeepScientist, EvoScientist), and a test on a second dataset before we call anything a gain.

Code, numbers and the full log: github.com/jarjlol/diffullama, branch `assignment/codification`*.*